

Mark Scheme

Q1.

	Scheme	Marks	AO
(a)	{Let X = time spent, $P(X > 15) = \}$ 0.105649... awrt 0.106	B1 (1)	1.1b
(b)	$H_0: \mu = 10$ $H_1: \mu > 10$ $\bar{X} \sim N\left(10, \left(\frac{4}{\sqrt{20}}\right)^2\right)$; $P(\bar{X} > 11.5) = 0.046766...$ [Condone 0.9532...] [This is significant ($< 5\%$) so] there is evidence to support the complaint	B1 M1;A1 A1 (4)	2.5 3.3;3.4 2.2b
(c)(i)	$P(T < 2) =$ 0.1956... awrt 0.196	B1 (1)	1.1b
(ii)	Require $\frac{P(0 < T < 2)}{P(T > 0)} = \frac{0.119119...}{0.923436...}$; = 0.1289955... awrt 0.129	M1 A1;A1 (3)	3.4 1.1bx2
(iii)	The current model suggests non-negligible probability of T values < 0 which is impossible	B1 (1)	3.5b
(d)	Require t such that $P(T > t T > 2) = 0.5$ or $P(T < t T > 2) = 0.5$ e.g. $\frac{P(T > t)}{P(T > 2)} = 0.5$; so $P(T > t) = 0.5 \times [1 - (c)(i)]$ or $P(T > t) = 0.5 \times 0.8043...$ [i.e. $P(T > t) = 0.40...$ implies] $\frac{t-5}{3.5} = 0.2533$ or $P(T < t) = "0.5978..."$ $t = 5.886...$ or from calculator 5.867... so awrt 5.9	M1 M1; A1ft M1 A1 (5)	3.1b 1.1b 3.4 1.1b 1.1b
		(15 marks)	

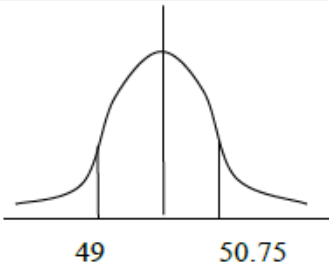
	Notes
(a)	B1 for awrt 0.106 (from calculator) [Allow 10.6%]
(b)	B1 for both hypotheses correct in terms of μ . M1 for selection of a correct model (sight or use of correct normal- may not have label $\bar{X}$) 1 st A1 for use of this model to get probability allow 0.046~0.047 [Condone awrt 0.953] ALT OR test statistic $z = 1.677...$ (awrt 1.68) and cv of 1.64 (or better) or CR $\bar{X} > 11.47..$ 2 nd A1 (dep on 1 st A1 or at least $P(\bar{X} > 11.5) < 0.05$ (o.e.)) for a correct conclusion in context -must mention complaint/claim or time/mins is > 10 SC (M0 for $\bar{X} \sim N(11.5, ...)$ for correct probability and conclusion (score M0A0A1 on open)
(c)(i)	B1 for awrt 0.196 (from calculator) [Allow 19.6%]
(ii)	M1 for a correct probability ratio expression (may be implied by 1 st A1 scored) 1 st A1 for a correct ratio of probabilities (both correct or truncated to 2 dp) 2 nd A1 for awrt 0.129
(iii)	B1 for a suitable explanation of why model is not suitable based on negative T values Must say that a significant proportion of values < 0 (o.e.) e.g. $P(T > 0)$ should be closer to 1 or Difference between $P(T < 2 T > 0)$ and $P(T < 2)$ is too big (o.e.)
(d)	1 st M1 for a correct conditional probability statement to start the problem or $0.5 \times P(T > 2)$ 2 nd M1 for correct ratio of probability expressions [Must have $P(T > t)$ or $P(2 < T < t)$] 1 st A1ft for a correct equation for $P(T > t)$ (o.e.) ft their answer to part (c) [May be in a diagram] 3 rd M1 for attempt to find t (standardising and sight of 0.2533) or prepare to use calc (ft) Arriving at $P(T < \text{median}) = 1 - 0.5 \times \text{their } 0.8043$ will score 1 st 4 marks 2 nd A1 for awrt 5.9 Sight of awrt 5.9 and at least one M mark scores 5/5 [Answer only send to review]

Q2.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	$\frac{24.63 - 25}{\sigma} = -1.0364$	M1	This mark is given for standardising as part of a method to find σ
	$\sigma = 0.357$	A1	This mark is given for a correct value of σ
	$P(D > K) = 0.4$ or $P(D < K) = 0.6$	B1	This mark is given for
	$\frac{k - 25}{\sigma} = \frac{k - 25}{0.357} = 0.2533$	M1	This mark is given for using a normal model to find the probability
	$k = 25.09$	A1	This mark is given for a correct value for k
(b)	$Y \sim B(200, 0.45)$ so $W \sim N(90, 49.5)$	B1	This mark is given for setting up the normal distribution approximation of the binomial
	$P(Y < 100) \approx P(W < 99.5) = P\left(Z < \frac{99.5 - 90}{\sqrt{49.5}}\right)$	M1	This mark is given for using the normal model with a continuity correction
	$= 0.912$	A1	This mark is given for finding a correct value of the probability
(c)	$H_0 : \mu = 25$ $H_1 : \mu < 25$	B1	This mark is given for both hypotheses in terms of μ found correctly
	$\bar{D} \sim N\left(25, \frac{0.16^2}{20}\right)$	M1	This mark is given for a method to set up the normal distribution
	$P(\bar{D} < 24.94) = 0.0468$	A1	This mark is given for using the model to find a correct p -value
	$p = 0.0468 < 0.05$, so reject H_0	M1	This mark is given for a correct comparison and non-contextual conclusion
	There is sufficient evidence to support Hannah's belief	A1	This mark is given for a correct conclusion in context stated
(Total 13 marks)			



Q3.

Question	Scheme	Marks	AOs
(a)			
	$P(L > 50.98) = 0.025$	B1cao	3.4
	$\therefore \frac{50.98 - \mu}{0.5} = 1.96$	M1	1.1b
	$\therefore \mu = 50$	A1cao	1.1b
	$P(49 < L < 50.75)$	M1	3.4
	$= 0.9104\dots$ awrt <u>0.910</u>	A1ft	1.1b
	(5)		
(b)	$S = \text{number of strips that cannot be used so } S \sim B(10, 0.090)$	M1	3.3
	$= P(S \leq 3) = 0.991166\dots$ awrt 0.991	A1	1.1b
	(2)		
(c)	$H_0: \mu = 50.1$ $H_1: \mu > 50.1$	B1	2.5
	$\bar{X} \sim N\left(50.1, \frac{0.6^2}{15}\right)$ and $\bar{X} > 50.4$	M1	3.3
	$P(\bar{X} > 50.4) = 0.0264$	A1	3.4
	$p = 0.0264 > 0.01$ or $z = 1.936\dots < 2.3263$ and not significant	A1	1.1b
	There is insufficient evidence that the <u>mean length</u> of strips is <u>greater than 50.1</u>	A1	2.2b
	(5)		

(12 marks)

Notes:

(a)

1st M1: for standardizing with μ and 0.5 and setting equal to a z value ($|z| > 1$)

2nd M1: for attempting the correct probability for strips that can be used

2nd A1ft: awrt 0.910 (allow ft of their μ)

(b)

M1: for identifying a suitable binomial distribution

A1: awrt 0.991 (from calculator)

(c)

B1: hypotheses stated correctly

M1: for selecting a correct model (stated or implied)

1st A1: for use of the correct model to find $p = \text{awrt } 0.0264$ (allow $z = \text{awrt } 1.94$)

2nd A1: for a correct calculation, comparison and correct statement

3rd A1: for a correct conclusion in context mentioning "mean length" and 50.1

Q4.

Question	Scheme	Marks	AOs
(a)	The seeds would be destroyed in the process so they would have none to sell	B1	2.4
		(1)	
(b)	$[S = \text{no. of seeds out of 24 that germinate, } S \sim B(24, 0.55)]$		
	$T = \text{no. of trays with at least 15 germinating. } T \sim B(10, p)$	M1	3.3
	$p = P(S \geq 15) = 0.299126\dots$	A1	1.1b
	So $P(T \geq 5) = 0.1487\dots$ awrt <u>0.149</u>	A1	1.1b
		(3)	
(c)	n is large and p close to 0.5	B1	1.2
		(1)	
(d)	$X \sim N(132, 59.4)$	B1	3.4
	$P(X \geq 149.5) = P\left(Z \geq \frac{149.5 - 132}{\sqrt{59.4}}\right)$	M1	1.1b
	$= 0.01158\dots$ awrt <u>0.0116</u>	A1cso	1.1b
		(3)	
(e)	e.g The probability is very small therefore there is evidence that the company's claim is incorrect.	B1	2.2b
		(1)	
(9 marks)			

Notes:
(a) B1: cao
(b) M1: for selection of an appropriate model for T 1 st A1: for a correct value of the parameter p (accept 0.3 or better) 2 nd A1: for awrt 0.149
(c) B1: both correct conditions
(d) B1: for correct normal distribution M1: for correct use of continuity correction A1: cso
(e) B1: correct statement

Q5.

Qu	Scheme	Marks	AO
(a)	$P(L > 16) = 0.69146...$ awrt 0.691	B1 (1)	1.1b
(b)	$P(L > 20 L > 16) = \frac{P(L > 20)}{P(L > 16)}$ $= \frac{0.308537...}{(a)} \text{ or } \frac{1-(a)}{(a)} = 0.44621...$ For calc to work require $(0.44621...)^4 = 0.03964...$ awrt 0.0396	M1 A1ft, A1 dM1 A1 (5)	3.1b 1.1b 1.1b 2.1 1.1b
(c)	Require: $[P(L > 4)]^2 \times [P(L > 20 L > 16)]^2$ $= (0.99976...)^2 \times ("0.44621...")^2$ $= 0.19901...$ awrt 0.199 (*)	M1 A1ft A1cso* (3)	1.1a 1.1b 1.1b
(d)	$H_0: \mu = 18 \quad H_1: \mu > 18$ $\bar{L} \sim N\left(18, \left(\frac{4}{\sqrt{20}}\right)^2\right)$ $P(\bar{L} > 19.2) = P(Z > 1.3416...) = 0.089856...$ (0.0899 > 5%) or (19.2 < 19.5) or 1.34 < 1.6449 so not significant Insufficient evidence to support Alice's claim (or belief)	B1 M1 A1 A1 A1 (5)	2.5 3.3 3.4 1.1b 3.5a
(14 marks)			

	Notes
(a)	B1 for evaluating probability using their calculator (awrt 0.691) Accept 0.6915
(b)	1 st M1 for a first step of identifying a suitable conditional probability (either form) 1 st A1ft for a ratio of probabilities with numerator = awrt 0.309 or 1 – (a) and denom = their (a) 2 nd A1 for awrt 0.446 (o.e.) Accept 0.4465 (from $\frac{0.3085}{0.691} = 0.44645...$) NB $\frac{P(16 < L < 20)}{P(L > 16)} = 0.5538...$ scores M1A1A1 when they do $1 - 0.5538 = 0.4462...$ 2 nd M1 (dep on 1 st M1) for 2 nd correct step i.e. (their 0.446...) ⁴ or $X \sim B(4, "0.446")$ and $P(X = 4)$ 3 rd A1 for awrt 0.0396
(c)	1 st M1 for a correct approach to solving the problem (May be implied by A1ft) 1 st A1ft for $P(L > 4) =$ awrt 0.9998 used and ft their 0.44621 in correct expression If use $P(L > 20) = 0.3085..$ as 0.446.. in (b) then M1 for $(0.3085..)^2 \times [P(L > 4)]^2$; A1ft as above * 2 nd A1cso for 0.199 or better with clear evidence of M1 [NB $(0.4662..)^2 = 0.199...$ is M0A0A0] Must see M1 scored by correct expression in symbols or values (M1A1ft)
(d)	B1 for both hypotheses in terms of μ M1 for selecting a suitable model. Sight of <u>normal</u> , <u>mean</u> 18, <u>sd</u> $\frac{4}{\sqrt{20}}$ (o.e.) or <u>variance</u> = 0.8 1 st A1 for using the model correctly. Allow awrt 0.0899 or 0.09 from correct prob. statement CR $(\bar{L}) > 19.471...$ (accept awrt 19.5) or CV of 1.6449 (or better: calc 1.6448536..)
ALT	2 nd A1 for correct non-contextual conclusion. Wrong comparison or contradictions A0 Error giving 2 nd A0 implies 3 rd A0 but just a correct contextual conclusion can score A1A1 3 rd A1 dep on M1 and 1 st A1 for a correct contextual conclusion mentioning <u>Alice's claim</u> / <u>belief</u> or <u>there is insufficient evidence that the mean lifetime is more than 18 hours</u>

Q6.

Question	Scheme	Marks	AOs
(a)	$H_0: p = 0.1 \quad H_1: p \neq 0.1$	B1	2.5
		(1)	
(b)	Use of $X \sim B(50, 0.1)$ implied by sight of one of awrt 0.0052 or awrt 0.9755 or awrt 0.0245	M1	3.4
	Critical regions $X = 0$ or $X \geq 10$	A1	1.1b
	$X = 0$ and $X \geq 10$ plus $P(X = 0) = \text{awrt } 0.0052$ and $P(X \geq 10) = \text{awrt } 0.0245$	A1	1.1b
	SC: Both CR correct with no probabilities and no distribution seen scores M0A1A0		
		(3)	
(c)	0.0297	B1ft	1.1b
		(1)	
(d)	15 is <u>in the critical region</u> therefore there is evidence to support the <u>manager's</u> belief	B1ft	2.2b
		(1)	
(6 marks)			

Notes		
(a)	B1	For both hypotheses in terms of p or π . Connected to H_0 and H_1 correctly Condone 10% but not 10
(b)	M1	Using correct distribution to find the probability associated with one tail of the CR If the correct distribution is <u>stated</u> (may be seen in part(a)) allow for one tail of the correct CR or one of (awrt 0.025 or awrt 0.005 or awrt 0.975) seen connected to a correct probability statement
	A1	Lower CR $X = 0 / X < 1 / X \leq 0$ [condone eg $P(X = 0)$ labelled as CR] Or Upper CR $X \geq 10$ or $X > 9$ [condone $P(X \geq 10)$ or labelled as CR]
	A1	Both CR's correct with the relevant probabilities Allow $\cup$ for "and" and $X > 9, X < 1, X \leq 0$ [do not allow $P(X = 0)$ or $P(X \geq 10)$ or] Allow CR in different form eg $(9, \infty), [10, \infty)$
(c)	B1ft	awrt 0.0297 or 2.97% or ft for the sum of the probabilities in (b) for "their 2 critical regions" if seen. If none seen it must be awrt 0.0297 SC M0 in (b) for a one tail test Allow B1ft for their one tail CR in (b) eg 0.0338 or 0.0245 or 0.0579
(d)	B1ft	A correct statement about 15 and "their CR" or sight $P(X \geq 15) = 0.0000738...$ and comparison with "their 0.0245" and a compatible correct statement in context. eg There is evidence that there has been a change in the <u>proportion/probability</u> arriving <u>late</u> Condone increase rather than change Do not allow contradicting statements. NB No CR given in (b) then B0

	Scheme	Marks	AO
(a)	$\left[\text{Let } F \sim N(166.5, 6.1^2) \right] P(F < k) = 0.01 \Rightarrow \frac{k-166.5}{6.1} = -2.3263$ $k = 152.309... \quad \underline{152} \text{ or awrt } \underline{152.3}$	M1 A1	3.4 1.1b
(b)	$[P(150 < F < 175) =] \quad 0.914840... \quad \text{awrt } \underline{0.915}$	B1	1.1b
(c)	$P(F > 160 150 < F < 175)$ $= \frac{P(160 < F < 175)}{P(150 < F < 175)} \text{ or } \frac{P(160 < F < 175)}{"(b)"} = \frac{0.7749487...}{"0.91484..."} = 0.84708... \text{ awrt } \underline{0.847}$	M1 M1 A1ft A1	3.1b 1.1b 1.1b 1.1b
(d)	$H_0: \mu = 166.5 \quad H_1: \mu < 166.5$ $[\text{Let } X = \text{height of female from 2}^{\text{nd}} \text{ country}] \quad \bar{X} \sim N\left(166.5, \left(\frac{7.4}{\sqrt{50}}\right)^2\right)$ $P(\bar{X} < 164.6) = 0.03472...$ $[0.0347... < 0.05 \text{ so significant or reject } H_0]$ There is evidence to support Mia's belief	B1 M1 A1 dA1	2.5 3.3 3.4 2.2b
		(4)	
		(11 marks)	

	Notes
(a)	M1 for standardising (allow $\pm$) with k , 166.5 and 6.1 and set equal to a z value $2.3 < z < 2.4$ A1 for 152 or awrt 152.3 Ans only 2/2 [Condone poor use of notation e.g. $P(\frac{k-166.5}{6.1}) = -2.3263$] Allow percentages instead of probabilities throughout.
(b)	B1 for awrt 0.915
(c)	1 st M1 for interpreting demand as an appropriate conditional probability ($\Rightarrow$ by 2 nd M1) 2 nd M1 for correct ratio of expressions (can fit their (b) on denominator) ($\Rightarrow$ by 1 st A1ft) 1 st A1ft for a correct ratio of probs (can fit their "0.9148..." to 3sf from (b) if > 0.775) 2 nd A1 for awrt 0.847
(d)	B1 for both correct hypotheses in terms of μ 1 st M1 for selecting the correct model (needn't use $\bar{X} \Rightarrow$ by standardisation or 1 st A1) 1 st A1 for correct use of the correct model i.e. awrt 0.035 (allow 0.04 if $P("X" < 164.6)$ seen) Condone $P("X" > 164.6) = 0.9652$ or awrt 0.97 only if comparison with 0.95 is made
ALT	Use of z value: Need to see $Z = -1.8(15...)$ and cv of ± 1.6449 (allow 1.64 or better) for 1 st A1
ALT	Use of CR or CV for $\bar{X}$: Need to see " $\bar{X}$ " $< 164.7786...$ or CV = ... (awrt 164.8) for 1 st A1 Condone truncation i.e 164.7 or better 2 nd dA1 (dep on M1A1 only) for a correct inference in context. Must mention <u>Mia's belief</u> or <u>mean height of females/women</u> Do NOT award if contradictory statements about hypotheses made e.g. "not sig"
SC	M0 for $\bar{X} \sim N(164.6, ...)$ If they achieve $p =$ awrt 0.035 (o.e. with z -value or CV of 166.3) and a correct conclusion in context is given score M0A0A1 [and SC for awrt 0.97 > 0.95 case]

	Scheme	Marks	AO
(a)	Negative	B1 (1)	1.2
(b)	Marc's suggestion is <u>compatible</u> because it's <u>negative correlation</u>	B1 (1)	2.4
(c)	$(r =) -0.54458266...$ awrt <u>-0.545</u>	B1 (1)	1.1b
(d)	$H_0: \rho = 0$ $H_1: \rho < 0$ [5% 1-tail cv =] $(\pm) 0.4259$ (significant result / reject H_0) There is evidence of negative <u>correlation</u> between the <u>number of letters</u> in (or <u>length</u> of) a student's last <u>name</u> and their first <u>name</u>	B1 M1 A1 (3)	2.5 1.1a 2.2b
		(6 marks)	

	Notes
(a)	B1 for "negative" Allow "slight" or "weak" etc Allow a description e.g. "as x increases y decreases" or in context e.g. "people with longer last names tend to have shorter first names" A comment of "negative skew" is B0 Need to see distinct or separate responses for (a) and (b)
(b)	B1 for a comment that suggests data is compatible with the suggestion and a suitable reason such as "there is negative correlation" <u>or</u> a description in x and y or in context <u>or</u> the points lie close to a line with <u>negative gradient</u> <u>or</u> draw line $y = x$ and state that <u>more points below the line</u> so <u>supports (or is compatible with)</u> his suggestion A reason based on just a single point is B0 e.g. "11 letters in last name has only 5 in first name"
(c)	B1 for awrt -0.545
(d)	B1 for both hypotheses correct in terms of ρ M1 for a critical value compatible with their H_1 : 1-tail: awrt ± 0.426 (condone ± 0.425) or 2-tail (B0 scored for H_1): awrt ± 0.497 If hypotheses are in words and can deduce whether one or two-tail then use their words. If no hypotheses or their H_1 is not clearly one or two tail assume one-tail A1 for compatible signs between cv and r and a correct conclusion in context mentioning <u>correlation</u> and <u>number of letters</u> or <u>length</u> and <u>name</u> (ft their value from (c)) Do NOT award this A mark if contradictory comments or working seen e.g. "accept H_0 " or comparison of 0.426 with significance level of 0.05 etc NB The M1A1 can be scored independently of the hypotheses